

Chapter 7 & 8

Rotational Motion About a Fixed Axis

- Torque, Rotational Inertia and Law
- Angular Momentum and Conservation of Angular Momentum



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In the study of **linear motion**, objects were treated as point particles without structure. It didn't matter where a force was applied, only whether it was applied or not.

Rotational motion is an important part of everyday life. The rotation of the Earth creates the cycle of day and night, the rotation of wheels enables easy vehicular motion.

Rotational motion, when combined with Newton's law of universal gravitation and his laws of motion, can also explain certain facts about space travel and satellite motion, such as where to place a satellite so it will remain fixed in position over the same spot on the Earth.



Rigid Object

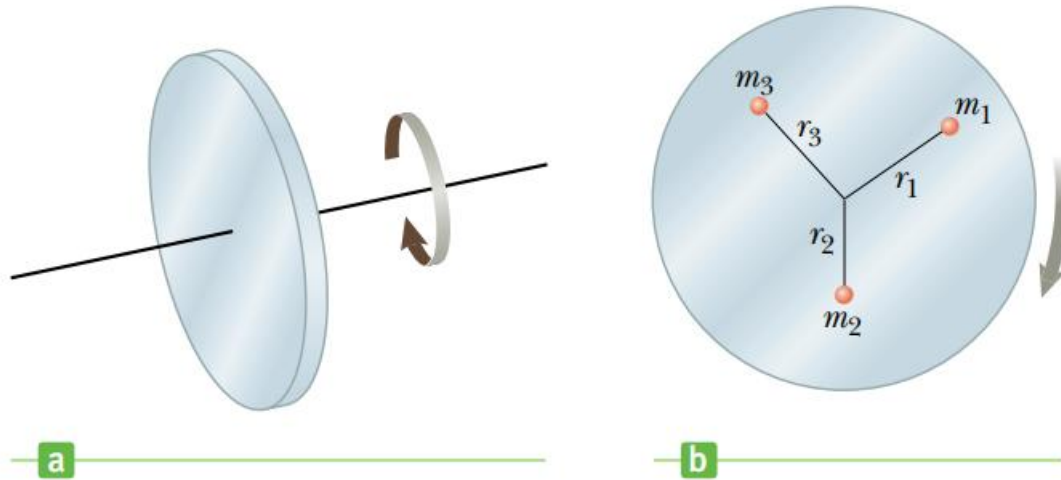


Figure 8.17 (a) A solid disk rotating about its axis. (b) The disk consists of many particles, all with the same angular acceleration.

- A rigid object is one that is **nondeformable**
 - The relative locations of all particles making up the object remain constant
 - All real objects are deformable to some extent, but the rigid object model is very useful in many situations where the deformation is negligible

7.1 Angular Quantities (P198)

(The point moves in circle with counter-clockwise direction)

Angular displacement: $\Delta\theta$

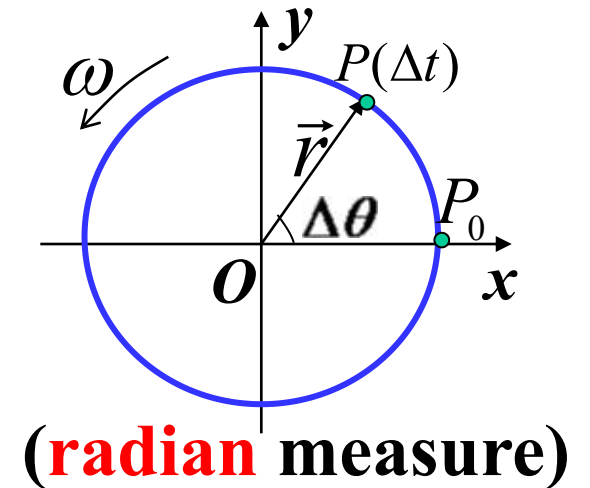
Angular velocity : $\omega = \lim_{\Delta t \rightarrow 0} \frac{\Delta\theta}{\Delta t} = \frac{d\theta}{dt}$

($\Delta\theta$ is the angle through which the point has rotated in the time Δt)

(ω equals θ for the first derivative of time)

Angular acceleration :

$$\alpha = \lim_{\Delta t \rightarrow 0} \frac{\Delta\omega}{\Delta t} = \frac{d\omega}{dt}$$



7.3 Relations between angular and linear quantities (P203)

1. The distance along a circular arc:

Assume the object rotates through the angle $\Delta\theta$, and hence point P moves through the arc length Δs , in the interval Δt . We know from the defining equation for radian measure that

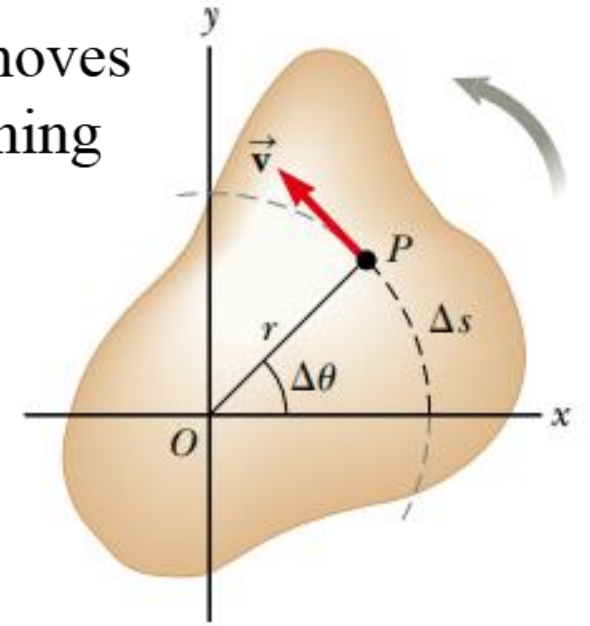
$$\Delta\theta = \frac{\Delta s}{r}$$

2. The linear speed (with r held constant):

(v equals s for the first derivative of time)

$$v = \frac{ds}{dt} = \frac{d\theta}{dt} r = \omega r$$

$$\boxed{v = \omega r}$$



Active Figure 7.5

Rotation of an object about an axis through O (the z -axis) that is perpendicular to the plane of the figure. Note that a point P on the object rotates in a circle of radius r centered at O .

3.The tangential component a_t :

Differentiating above equation with respect to time (r again holds constant)

$$\frac{dv}{dt} = \frac{d\omega}{dt} r$$

The tangential component a_t : $\alpha_t = \alpha r$

The tangential acceleration of a point on a rotating object equals the distance of that point from the axis of rotation multiplied by the angular acceleration.

4.Centripetal acceleration a_c (Normal acceleration a_n) :

$$a_n = \frac{v^2}{r} = r \frac{v^2}{r^2} = r \omega^2$$

8.1 Torque (P238)

1. General definition of torque

Let $\vec{F}$ be a force acting on an object, and let $\vec{r}$ be a position vector from a chosen point O to the point of application of the force. the torque $\vec{\tau}$ exerted by the force $\vec{F}$ is $\boxed{\vec{\tau} = \vec{r} \times \vec{F}}$

The magnitude of the torque $\tau = rF \sin \theta = Fd$

The direction of the torque: determined by **right-hand rule**

where **d** is called **moment arm** —the distance from the axis of rotation that is **perpendicular** to the line of action of the force.

Its **SI units**: **m · N**

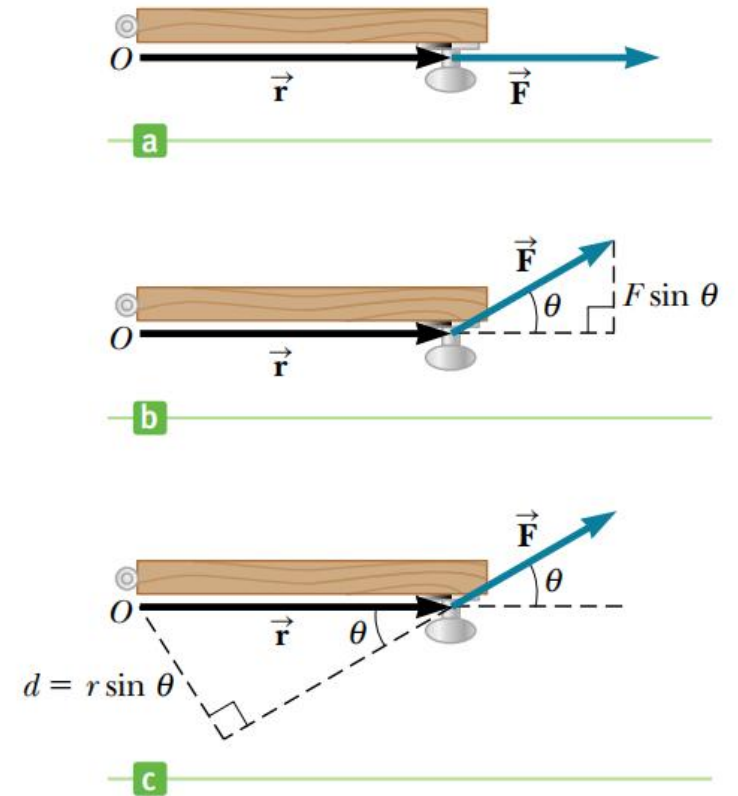


Figure 8.4 (a) A force $\vec{F}$ acting at an angle $\theta = 0^\circ$ exerts zero torque about the pivot O. (b) The part of the force perpendicular to the door, $F \sin \theta$, exerts torque $rF \sin \theta$ about O. (c) An alternate interpretation of torque in terms of a *lever arm* $d = r \sin \theta$.

EXAMPLE 8.2

The Swinging Door

GOAL Apply the more general definition of torque.

PROBLEM (a) A man applies a force of $F = 3.00 \times 10^2 \text{ N}$ at an angle of 60.0° to the door of Figure 8.7a, 2.00 m from the hinges. Find the torque on the door, choosing the position of the hinges as the axis of rotation. (b) Suppose a wedge is placed 1.50 m from the hinges on the other side of the door. What minimum force must the wedge exert so that the force applied in part (a) won't open the door?

STRATEGY Part (a) can be solved by substitution into the general torque equation. In part (b) the hinges, the wedge, and the applied force all exert torques on the door. The door doesn't open, so the sum of these torques must be zero, a condition that can be used to find the wedge force.

SOLUTION

(a) Compute the torque due to the applied force exerted at 60.0° .

Substitute into the general torque equation:

(b) Calculate the force exerted by the wedge on the other side of the door.

Set the sum of the torques equal to zero:

The hinge force provides no torque because it acts at the axis ($r = 0$). The wedge force acts at an angle of -90.0° , opposite the upward 260 N component.

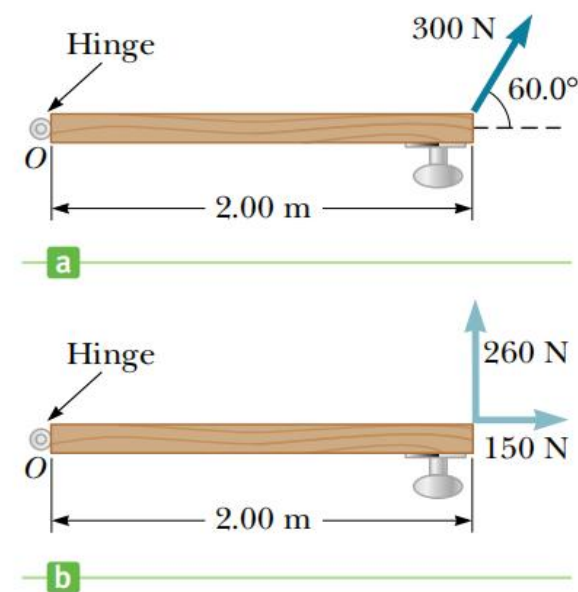


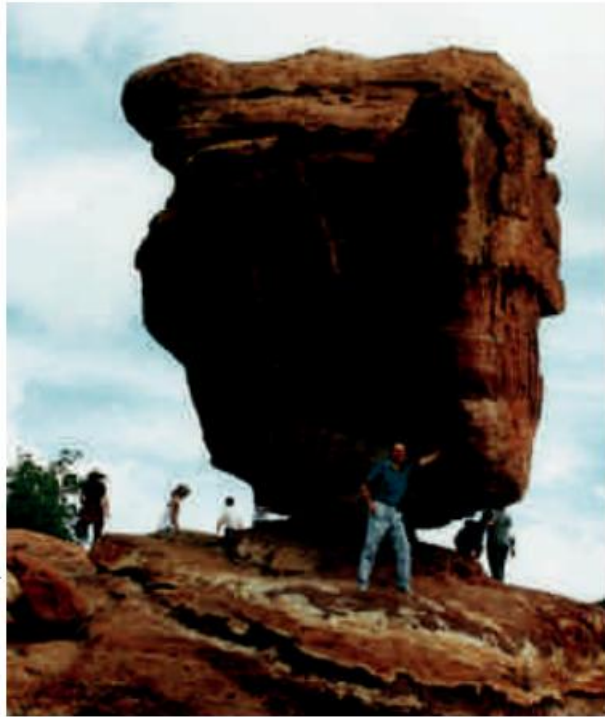
Figure 8.7 (Example 8.2a)
(a) Top view of a door being pushed by a 300-N force.
(b) The components of the 300-N force.

$$\begin{aligned}\tau_F &= rF \sin \theta = (2.00 \text{ m})(3.00 \times 10^2 \text{ N}) \sin 60.0^\circ \\ &= (2.00 \text{ m})(2.60 \times 10^2 \text{ N}) = 5.20 \times 10^2 \text{ N} \cdot \text{m}\end{aligned}$$

$$\tau_{\text{hinge}} + \tau_{\text{wedge}} + \tau_F = 0$$

$$\begin{aligned}0 + F_{\text{wedge}}(1.50 \text{ m}) \sin (-90.0^\circ) + 5.20 \times 10^2 \text{ N} \cdot \text{m} &= 0 \\ F_{\text{wedge}} &= 347 \text{ N}\end{aligned}$$

8.2 Torque and the Two Conditions for Equilibrium (P240)



David Serway

This large balanced rock at the Garden of the Gods in Colorado Springs, Colorado, is in mechanical equilibrium.

An object in mechanical equilibrium must satisfy the following two conditions:

1. The net external force must be zero: $\sum \vec{F} = 0$
2. The net external torque must be zero: $\sum \vec{\tau} = 0$

The first condition is a statement of translational equilibrium: The sum of all forces acting on the object must be zero, so the object has no translational acceleration, $\vec{a} = 0$. The second condition is a statement of rotational equilibrium: The sum of all torques on the object must be zero, so the object has no angular acceleration, $\vec{\alpha} = 0$. For an object to be in equilibrium, it must move through space at a constant speed and rotate at a constant angular speed.

EXAMPLE 8.3 Balancing Act

GOAL Apply the conditions of equilibrium and illustrate the use of different axes for calculating the net torque on an object.

PROBLEM A woman of mass $m = 55.0$ kg sits on the left end of a seesaw—a plank of length $L = 4.00$ m, pivoted in the middle as in Figure 8.8. (a) First compute the torques on the seesaw about an axis that passes through the pivot point. Where should a man of mass $M = 75.0$ kg sit if the system (seesaw plus man and woman) is to be balanced? (b) Find the normal force exerted by the pivot if the plank has a mass of $m_{\text{pl}} = 12.0$ kg. (c) Repeat part (a), but this time compute the torques about an axis through the left end of the plank.

STRATEGY In part (a), apply the second condition of equilibrium, $\Sigma\tau = 0$, computing torques around the pivot point. The mass of the plank forming the seesaw is distributed evenly on either side of the pivot point, so the torque exerted by gravity on the plank, τ_{gravity} , can be computed as if all the plank's mass is concentrated at the pivot point. Then τ_{gravity} is zero, as is the torque exerted by the pivot, because their lever arms are zero. In part (b) the first condition of equilibrium, $\Sigma\vec{F} = 0$, must be applied. Part (c) is a repeat of part (a) showing that choice of a different axis yields the same answer.

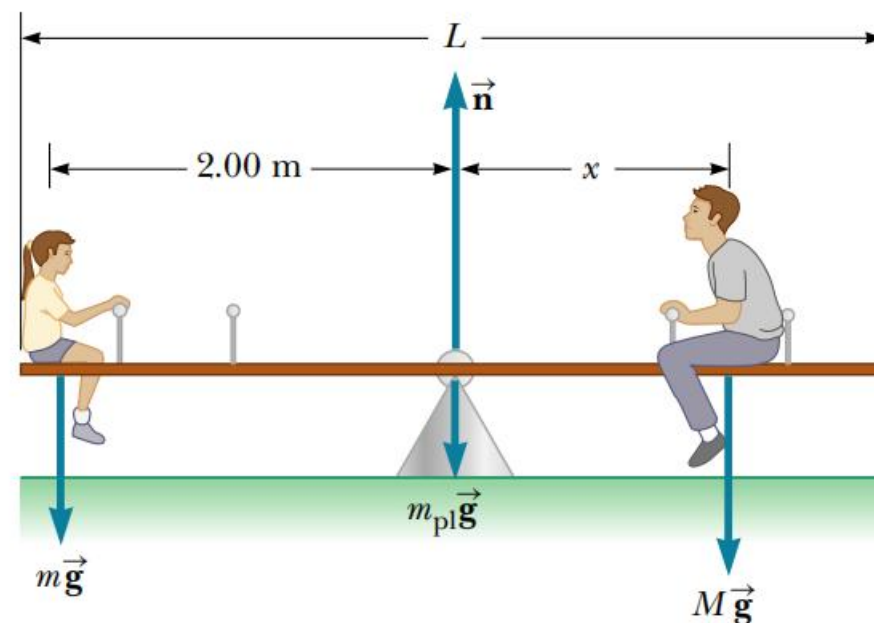


Figure 8.8 (Example 8.3) The system consists of two people and a seesaw. Because the sum of the forces and the sum of the torques acting on the system are both zero, the system is said to be in equilibrium.

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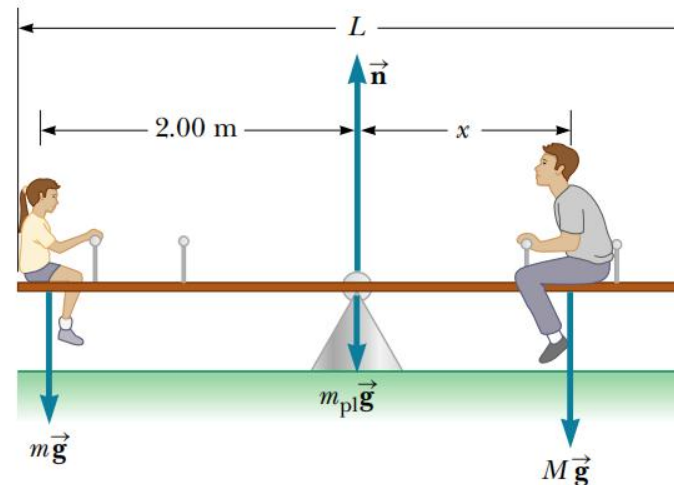


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SOLUTION

(a) Where should the man sit to balance the seesaw?

Apply the second condition of equilibrium to the plank by setting the sum of the torques equal to zero:

The first two torques are zero. Let x represent the man's distance from the pivot. The woman is at a distance $L/2$ from the pivot.

Solve this equation for x and evaluate it:

$$\tau_{\text{pivot}} + \tau_{\text{gravity}} + \tau_{\text{man}} + \tau_{\text{woman}} = 0$$

$$0 + 0 - Mgx + mg(L/2) = 0$$

$$x = \frac{m(L/2)}{M} = \frac{(55.0 \text{ kg})(2.00 \text{ m})}{75.0 \text{ kg}} = 1.47 \text{ m}$$

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SOLUTION

(b) Find the normal force n exerted by the pivot on the seesaw.

Apply for first condition of equilibrium to the plank, solving the resulting equation for the unknown normal force, n :

$$\begin{aligned}-Mg - mg - m_{\text{pl}}g + n &= 0 \\ n &= (M + m + m_{\text{pl}})g \\ &= (75.0 \text{ kg} + 55.0 \text{ kg} + 12.0 \text{ kg})(9.80 \text{ m/s}^2) \\ n &= 1.39 \times 10^3 \text{ N}\end{aligned}$$

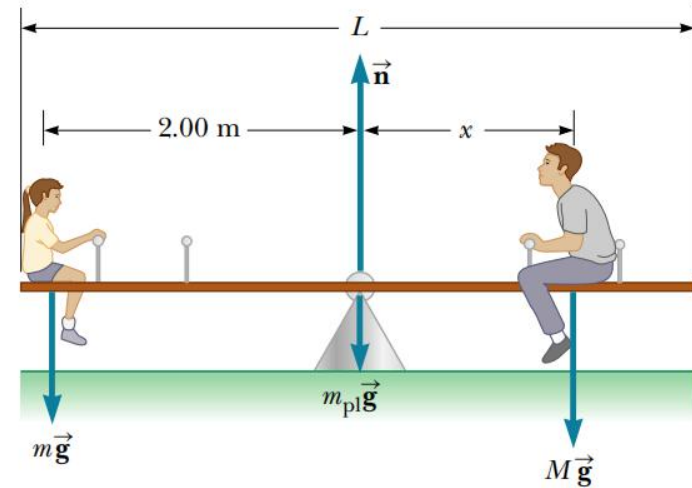


Figure 8.8 (Example 8.3) The system consists of two people and a seesaw. Because the sum of the forces and the sum of the torques acting on the system are both zero, the system is said to be in equilibrium.

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SOLUTION

(c) Repeat part (a), choosing a new axis through the left end of the plank.

Compute the torques using this axis, and set their sum equal to zero. Now the pivot and gravity forces on the plank result in nonzero torques.

Substitute all known quantities:

Solve for x , substituting the normal force found in part (b):

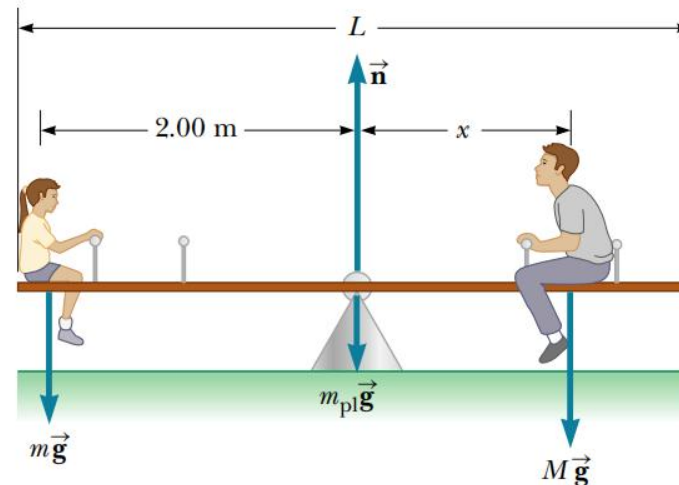


Figure 8.8 (Example 8.3) The system consists of two people and a seesaw. Because the sum of the forces and the sum of the torques acting on the system are both zero, the system is said to be in equilibrium.

$$\begin{aligned}
 \tau_{\text{man}} + \tau_{\text{woman}} + \tau_{\text{plank}} + \tau_{\text{pivot}} &= 0 \\
 -Mg(L/2 + x) + mg(0) - m_{\text{pl}}g(L/2) + n(L/2) &= 0 \\
 -(75.0 \text{ kg})(9.80 \text{ m/s}^2)(2.00 \text{ m} + x) + 0 \\
 - (12.0 \text{ kg})(9.80 \text{ m/s}^2)(2.00 \text{ m}) + n(2.00 \text{ m}) &= 0 \\
 -(1.47 \times 10^3 \text{ N} \cdot \text{m}) - (735 \text{ N})x - (235 \text{ N} \cdot \text{m}) \\
 + (2.00 \text{ m})n &= 0
 \end{aligned}$$

$$x = 1.46 \text{ m}$$

8.5 Relationship Between Torque and Angular Acceleration(P247)

(N-II Law for Rotation)

$$\tau_{\text{net}} = I\alpha$$

I — the **moment of inertia** or **rotational inertia** on the axis.

—— Net torque on a rigid body equals to the product of rotational inertia and angular acceleration it causes **about same axis**.

This equation is **the rotational analog of Newton's second law of motion**, with torque replacing force, moment of inertia replacing mass, and angular acceleration replacing linear acceleration. The moment of inertia I is a measure of the **rotational inertia** of a body, which plays the same role for rotational motion that mass does for translational motion.

Moment of Inertia for Particle System (P248)

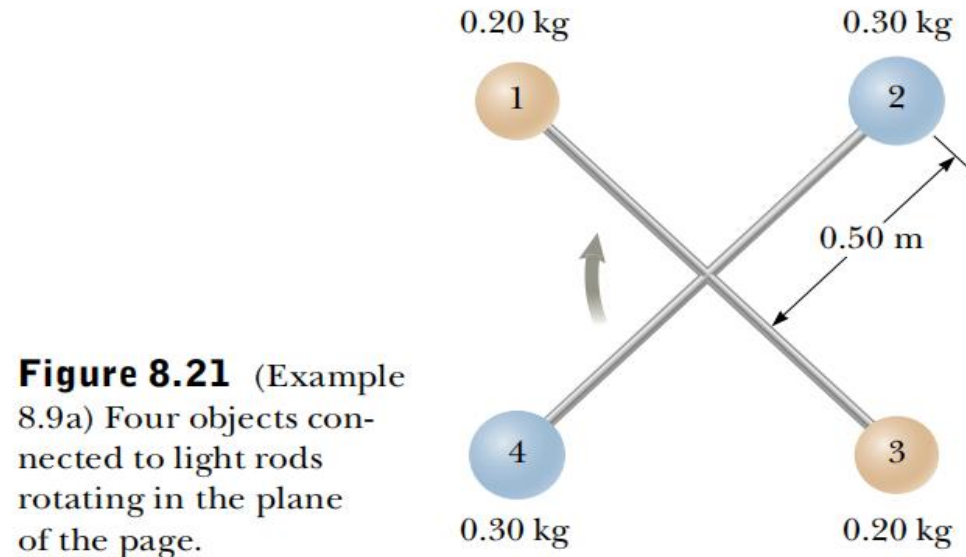


Figure 8.21 (Example 8.9a) Four objects connected to light rods rotating in the plane of the page.

If the masses and distances of the particles are labeled with subscripts as in Figure 8.21 , the moment of inertia, I , of the whole body is :

$$I \equiv \sum mr^2 = m_1 r_1^2 + m_2 r_2^2 + m_3 r_3^2 + \dots$$

EXAMPLE 8.9

The Baton Twirler

GOAL Calculate a moment of inertia.

PROBLEM In an effort to be the star of the halftime show, a majorette twirls an unusual baton made up of four spheres fastened to the ends of very light rods (Fig. 8.21). Each rod is 1.0 m long. **(a)** Find the moment of inertia of the baton about an axis perpendicular to the page and passing through the point where the rods cross. **(b)** The majorette tries spinning her strange baton about the axis OO' , as shown in Figure 8.22 on page 250. Calculate the moment of inertia of the baton about this axis.

SOLUTION

(a) Calculate the moment of inertia of the baton when oriented as in Figure 8.21.

Apply Equation 8.7, neglecting the mass of the connecting rods:

$$\begin{aligned} I &= \sum mr^2 = m_1 r_1^2 + m_2 r_2^2 + m_3 r_3^2 + m_4 r_4^2 \\ &= (0.20 \text{ kg})(0.50 \text{ m})^2 + (0.30 \text{ kg})(0.50 \text{ m})^2 \\ &\quad + (0.20 \text{ kg})(0.50 \text{ m})^2 + (0.30 \text{ kg})(0.50 \text{ m})^2 \\ I &= 0.25 \text{ kg} \cdot \text{m}^2 \end{aligned}$$

(b) Calculate the moment of inertia of the baton when oriented as in Figure 8.22.

Apply Equation 8.7 again, neglecting the radii of the 0.20-kg spheres.

$$\begin{aligned} I &= \sum mr^2 = m_1 r_1^2 + m_2 r_2^2 + m_3 r_3^2 + m_4 r_4^2 \\ &= (0.20 \text{ kg})(0)^2 + (0.30 \text{ kg})(0.50 \text{ m})^2 + (0.20 \text{ kg})(0)^2 \\ &\quad + (0.30 \text{ kg})(0.50 \text{ m})^2 \\ I &= 0.15 \text{ kg} \cdot \text{m}^2 \end{aligned}$$

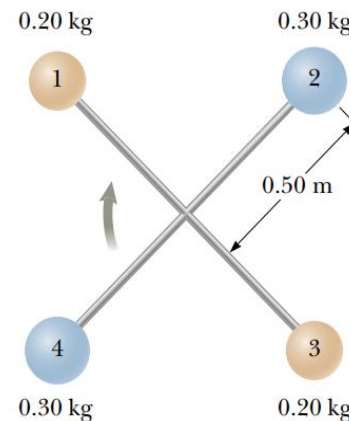


Figure 8.21 (Example 8.9a) Four objects connected to light rods rotating in the plane of the page.

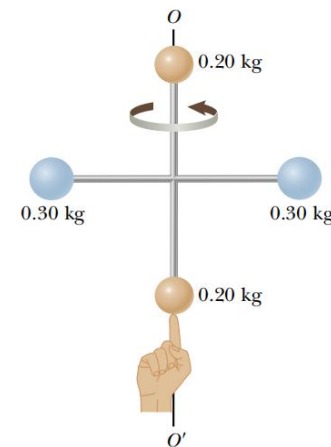


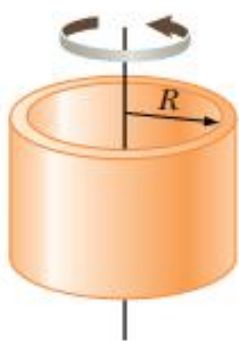
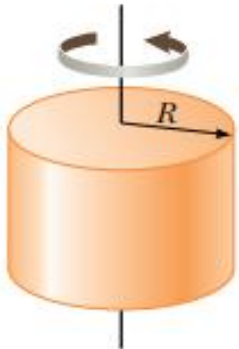
Figure 8.22 (Example 8.9b) A double baton rotating about the axis OO' .

Moments of Inertia for Extended Objects (P250)

Divided the extended objects into many small volume elements, each of mass Δm_i
We can rewrite the expression for I in terms of Δm

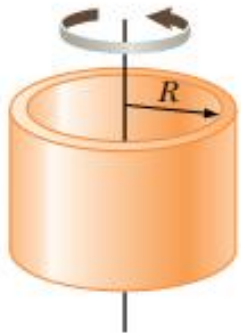
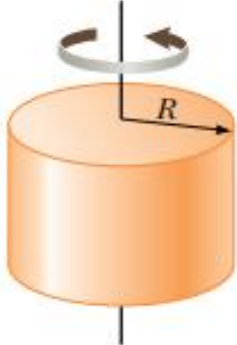
$$I = \lim_{\Delta m_i \rightarrow 0} \sum_i r_i^2 \Delta m_i = \int r^2 dm$$

Table 8.1 Moments of Inertia for Various Rigid Objects of Uniform Composition

Hoop or thin cylindrical shell $I = MR^2$		Solid cylinder or disk $I = \frac{1}{2}MR^2$		Long, thin rod with rotation axis through center $I = \frac{1}{12}ML^2$		Long, thin rod with rotation axis through end $I = \frac{1}{3}ML^2$	
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Moments of Inertia for Extended Objects (P250)

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Descriptions:

- 1、 Moments of Inertia is the **measure for the inertia** of rotation .($\text{kg}\cdot\text{m}^2$) In general, smaller I means easier rotation.
- 2、 It is determined by the total mass and the **distribution of mass** to the axis, so rotational inertia relative to **different axis is different** even for the same body.

The rotational inertia I is related to **mass**, **mass distribution** and **position of axis**。

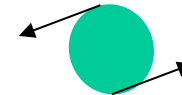
Strategy to use the Newton 2nd Law

- Draw or sketch system. Adopt coordinates, indicate rotation axes, list the known and unknown quantities, ...
- Draw free body diagrams of key parts. Show forces at their points of application. Find torques about a (common) axis
- May need to apply Second Law twice, once to each part

➤ **Translation:** $\vec{F}_{\text{net}} = \sum \vec{F}_i = m\vec{a}$

➤ **Rotation:** $\vec{\tau}_{\text{net}} = \sum \vec{\tau}_i = I\vec{\alpha}$

Note: can have
 $F_{\text{net}} = 0$
but $\tau_{\text{net}} \neq 0$



- The rotational inertia I is a measure of the rotational inertia of a body, which plays the same role for rotational motion that mass does for translational motion.
- Make sure there are enough (N) equations; there may be constraint equations (extra conditions connecting unknowns)
- Simplify and solve the set of (simultaneous) equations.
- Find unknown quantities and check answers

EXAMPLE 8.11 The Falling Bucket

GOAL Combine Newton's second law with its rotational analog.

PROBLEM A solid, uniform, frictionless cylindrical reel of mass $M = 3.00$ kg and radius $R = 0.400$ m is used to draw water from a well (Fig. 8.25a). A bucket of mass $m = 2.00$ kg is attached to a cord that is wrapped around the cylinder. (a) Find the tension T in the cord and acceleration a of the bucket.

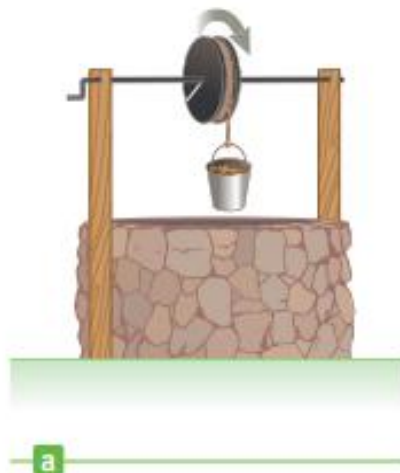
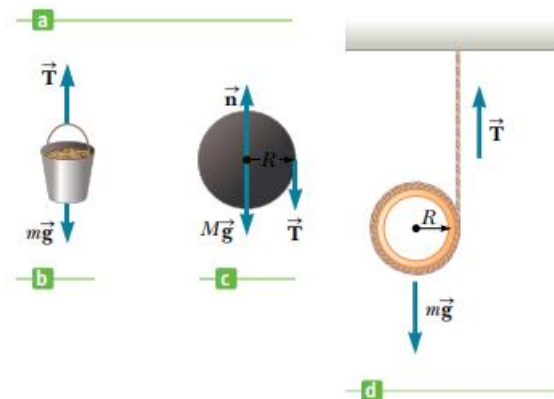


Figure 8.25 (Example 8.11) (a) A water bucket attached to a rope passing over a frictionless reel. (b) A force diagram for the bucket. (c) The tension produces a torque on the cylinder about its axis of rotation. (d) A falling cylinder (Exercise 8.11).



SOLUTION

(a) Find the tension in the cord and the acceleration of the bucket.

Apply Newton's second law to the bucket in Figure 8.25b. There are two forces: the tension $\vec{T}$ acting upward and gravity $m\vec{g}$ acting downward.

Apply $\tau = I\alpha$ to the cylinder in Figure 8.25c:

Notice the angular acceleration is clockwise, so the torque is negative. The normal and gravity forces have zero moment arm and don't contribute any torque.

Solve for T and substitute $\alpha = a/R$ (notice that both α and a are negative):

Substitute the expression for T in Equation (3) into Equation (1), and solve for the acceleration:

Substitute the values for m , M , and g , getting a , then substitute a into Equation (3) to get T :

$$(1) \quad ma = -mg + T$$

$$\sum \tau = I\alpha = \frac{1}{2}MR^2\alpha \quad (\text{solid cylinder})$$

$$(2) \quad -TR = \frac{1}{2}MR^2\alpha$$

$$(3) \quad T = -\frac{1}{2}MR\alpha = -\frac{1}{2}Ma$$

$$ma = -mg - \frac{1}{2}Ma \quad \rightarrow \quad a = -\frac{mg}{m + \frac{1}{2}M}$$

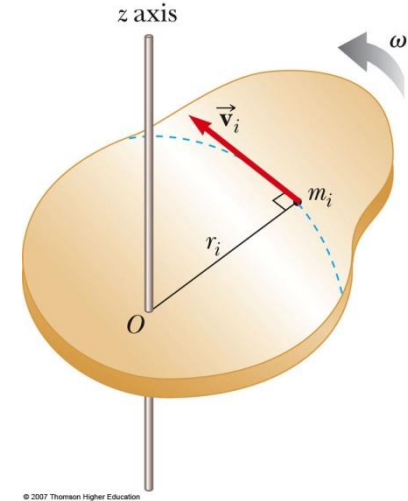
$$a = -5.60 \text{ m/s}^2 \quad T = 8.40 \text{ N}$$

8.6 Rotational Kinetic Energy (P254)

- An object rotating about z axis with an angular speed, ω , has rotational kinetic energy
- The total rotational kinetic energy of the rigid object is the sum of the energies of all its particles

$$K_R = \sum_i K_i = \sum_i \frac{1}{2} m_i r_i^2 \omega^2$$
$$K_R = \frac{1}{2} \left(\sum_i m_i r_i^2 \right) \omega^2 = \frac{1}{2} I \omega^2$$

- Where I is called the moment of inertia
- Unit of rotational kinetic energy is Joule (J)



Example:

ESTIMATE Flywheel. Flywheels, which are simply large rotating disks, have been suggested as a means of storing energy for solar-powered generating systems. Estimate the kinetic energy that can be stored in a 20,000-kg flywheel with a diameter of 20m (a six-story building). Assume it could hold together (without flying apart due to internal stresses) at 100 rpm.

Solution:

the rotational inertia of the disk is $I = \frac{1}{2}mR^2$

so the kinetic energy stored in the disk is

$$K = \frac{1}{2}I\omega^2 = \frac{1}{2}\left(\frac{1}{2}mR^2\right)\omega^2 = 5.2 \times 10^6 J$$

Work-kinetic energy theorem for fixed-axis rotation

From the kinetic theorem of a single particle, $W = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2 = \Delta K$

for a rotation $\longrightarrow$ $W = \frac{1}{2}I\omega_f^2 - \frac{1}{2}I\omega_i^2 = \Delta K$

where $W = \int_{\theta_i}^{\theta_f} \tau d\theta$

— the **work** done on a rigid body during a finite angular displacement **equals** to **the change of** its rotational kinetic energy.

Example

A centrifuge rotor has a moment of inertia of $4.25 \times 10^{-2} \text{ kg} \cdot \text{m}^2$. How much energy is required to bring it from rest to 10,000 rpm?

Solution

The work done increases the kinetic energy of the rotor:

$$\begin{aligned} W &= \Delta K = \frac{1}{2} I \omega^2 - 0 \\ &= \frac{1}{2} (4.25 \times 10^{-2} \text{ kg} \cdot \text{m}^2) [(10,000 \text{ rev/min})(2\pi \text{ rad/rev}) / (60 \text{ s/min})]^2 = \boxed{2.33 \times 10^4 \text{ J}.} \end{aligned}$$

Work-kinetic energy theorem for fixed-axis rotation

The principle of work-energy for particle-systems is still hold for the rigid bodies.

$$W_{\text{ext}} + W_{nc_i} = (K_f + U_f) - (K_i + U_i) = \Delta E_{\text{mec}}$$

$$\text{when } W_{\text{ext}} + W_{nc_i} = 0 \quad \Rightarrow \quad K_f + U_f = K_i + U_i$$

If **only** conservative torques do work, mechanical energy **conserves** —— both kinetic energy and potential energy transfer each other.

Work-kinetic energy theorem for fixed-axis rotation

- For pure translation

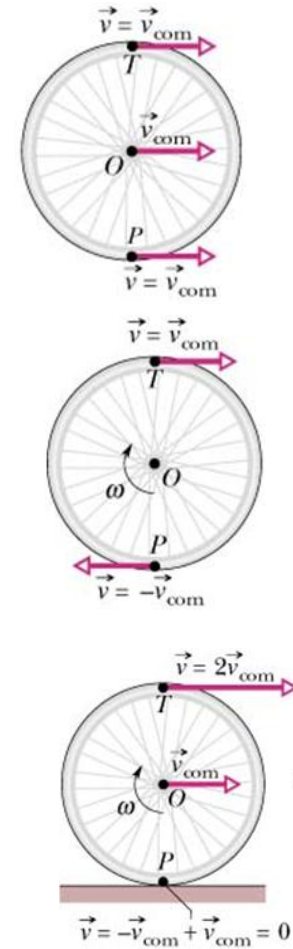
$$W_{net} = \Delta K_{cm} = K_{cm,f} - K_{cm,i} = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2$$

- For pure rotation

$$W_{net} = \Delta K_{rot} = K_{rot,f} - K_{rot,i} = \frac{1}{2}I\omega_f^2 - \frac{1}{2}I\omega_i^2$$

- Rolling: pure rotation + pure translation

$$\begin{aligned} W_{net} &= \Delta K_{total} = (K_{rot,f} + K_{cm,f}) - (K_{rot,i} + K_{cm,i}) \\ &= \left(\frac{1}{2}I\omega_f^2 + \frac{1}{2}mv_f^2 \right) - \left(\frac{1}{2}I\omega_i^2 + \frac{1}{2}mv_i^2 \right) \end{aligned}$$



Work-kinetic energy theorem for fixed-axis rotation

The principle of work-energy for particle-systems is still hold for the rigid bodies.

$$W_{\text{ext}} + W_{nc_i} = (K_f + U_f) - (K_i + U_i) = \Delta E_{\text{mec}}$$

(the work done by **external** torques and **nonconservative internal** torques = change of mechanical energy of a system)

$$\text{when } W_{\text{ext}} + W_{nc_i} = 0 \quad \Rightarrow \quad K_f + U_f = K_i + U_i$$

If **only** conservative torques do work, mechanical energy conserves —— both kinetic energy and potential energy transfer each other.

■ EXAMPLE 8.12 A Ball Rolling Down an Incline

GOAL Combine gravitational, translational, and rotational energy.

PROBLEM A ball of mass M and radius R starts from rest at a height of 2.00 m and rolls down a 30.0° slope, as in Figure 8.27. What is the linear speed of the ball when it leaves the incline? Assume that the ball rolls without slipping.

STRATEGY The two points of interest are the top and bottom of the incline, with the bottom acting as the zero point of gravitational potential energy. As the ball rolls down the ramp, gravitational potential energy is converted into both translational and rotational kinetic energy without dissipation, so conservation of mechanical energy can be applied with the use of Equation 8.11.

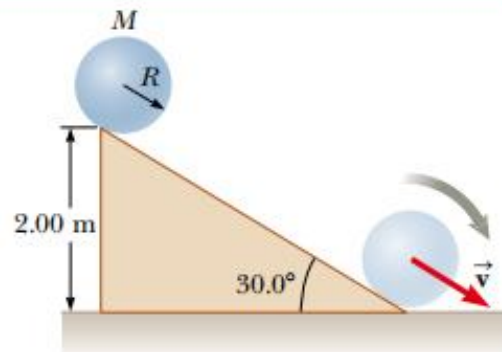


Figure 8.27 (Example 8.12) A ball starts from rest at the top of an incline and rolls to the bottom without slipping.

SOLUTION

Apply conservation of energy with $PE = PE_g$, the potential energy associated with gravity:

Substitute the appropriate general expressions, noting that $(KE_t)_i = (KE_r)_i = 0$ and $(PE_g)_f = 0$ (obtain the moment of inertia of a ball from Table 8.1):

The ball rolls without slipping, so $R\omega = v$, the “no-slip condition,” can be applied:

Solve for v , noting that M cancels.

$$(KE_t + KE_r + PE_g)_i = (KE_t + KE_r + PE_g)_f$$

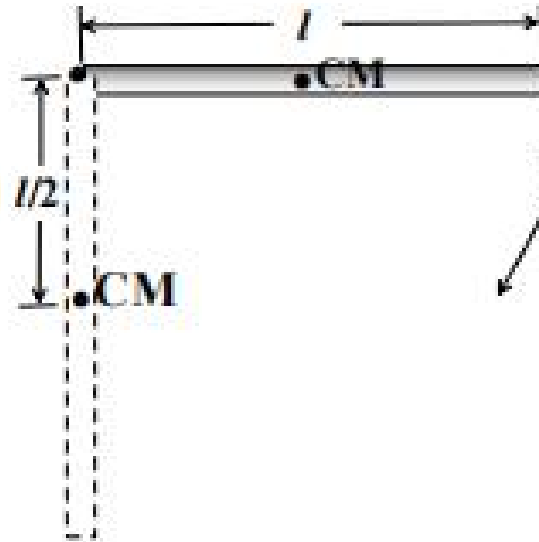
$$0 + 0 + Mgh = \frac{1}{2}Mv^2 + \frac{1}{2}\left(\frac{2}{5}MR^2\right)\omega^2 + 0$$

$$Mgh = \frac{1}{2}Mv^2 + \frac{1}{5}Mv^2 = \frac{7}{10}Mv^2$$

$$v = \sqrt{\frac{10gh}{7}} = \sqrt{\frac{10(9.80 \text{ m/s}^2)(2.00 \text{ m})}{7}} = 5.29 \text{ m/s}$$

Example

Rotating rod. A rod of mass M is pivoted on a frictionless hinge at one end, as shown in Fig. . The rod is held at rest horizontally and then released. Determine the angular velocity of the rod when it reaches the vertical position, and the speed of the rod's tip at this moment.

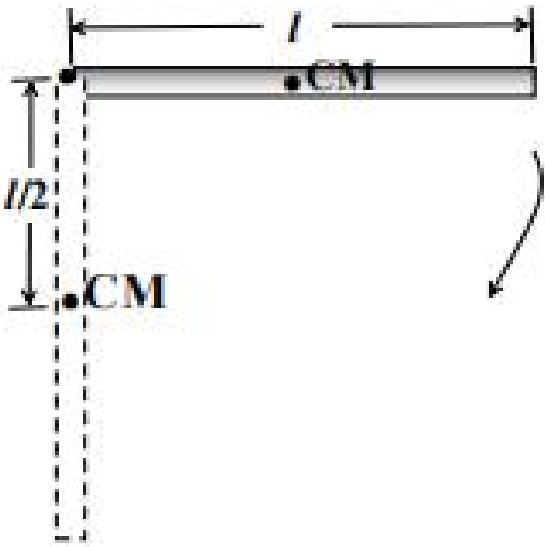


Fig

Solution

We can use the work-energy principle here. The work done is due to gravity, and is equal to the change in gravitational potential energy of the rod. Since the CM of the rod drops a vertical distance $l/2$, the work done by gravity is

$$W = Mg \frac{l}{2}$$



The initial kinetic energy is zero. Hence, from the work-energy principle,

$$\frac{1}{2} I \omega^2 = Mg \frac{l}{2}$$

Since $I = \frac{1}{3} ml^2$ for a rod pivoted about its end, we can solve for ω :

$$\omega = \sqrt{\frac{3g}{l}}$$

The tip of the rod will have a linear speed

$$v = l\omega = \sqrt{3gl}$$

By comparison, an object that falls vertically a height l has a speed $v = \sqrt{2gl}$

8.7 Angular Momentum (P257)

1. Angular momentum of a particle

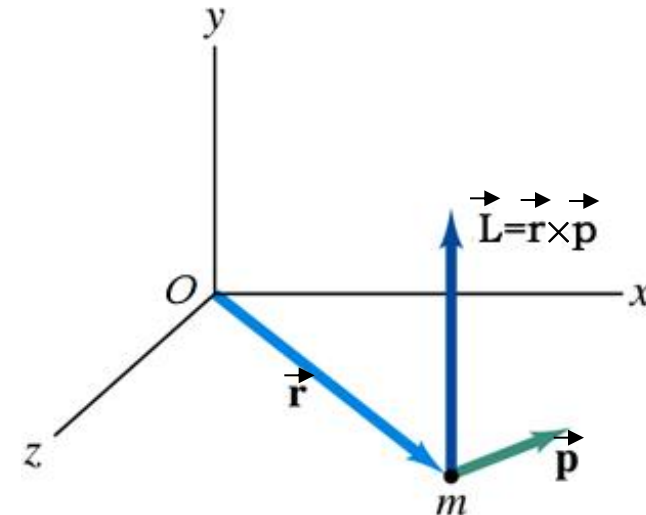
The linear momentum of a particle:

$$\vec{P} = m\vec{v}$$

The Angular momentum relative to a fixed point:

$$\vec{l} = \vec{r} \times \vec{p} = m(\vec{r} \times \vec{v})$$

Magnitude: $l = |\vec{l}| = mvr \sin \varphi$
Direction: right-hand rule

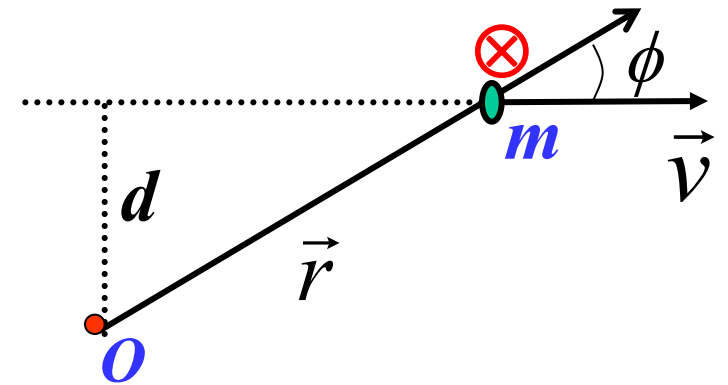


Supplementary explanation:

- (1) It is a vector quantity (SI unit: **J.s**);
- (2) It is **not same** relative to different point.
- (3) The angular momentum of particle relative to some point in the motion of straight line.

$$\vec{l} = \vec{r} \times \vec{p} = m(\vec{r} \times \vec{v})$$

$$\left\{ \begin{array}{l} \text{Magnitude: } l = |\vec{l}| = mvr \sin \phi = mvd \\ \text{Direction: right-hand rule } \otimes \end{array} \right.$$



d : the vertical distance between the point O and the line in the velocity direction.

Example

Determine the angular momentum of a 60-g particle about the origin of coordinates when the particle is at $x = 7.0$ m, $y = -6.0$ m, and it has velocity $\vec{v} = (2.0\vec{i} - 8.0\vec{k})$ m/s.

Solution

For the angular momentum we have

$$\begin{aligned}\vec{L} &= \vec{r} \times m\vec{v} = m(\vec{r} \times \vec{v}) \\ &= (0.060 \text{ kg})[(7.0 \text{ m})\vec{i} + (-6.0 \text{ m})\vec{j}] \times [(2.0 \text{ m/s})\vec{i} - (8.0 \text{ m/s})\vec{k}] \\ &= (0.060 \text{ kg})\{[(-6.0 \text{ m})(-8.0 \text{ m/s}) - 0]\vec{i} + [0 - (7.0 \text{ m})(-8.0 \text{ m/s})]\vec{j} + [0 - (-6.0 \text{ m})(2.0 \text{ m/s})]\vec{k}\} \\ &= \boxed{(2.9\vec{i} + 3.4\vec{j} + 0.72\vec{k}) \text{ kg} \cdot \text{m}^2/\text{s}.}\end{aligned}$$

2. The angular momentum of a system of particles

For a system of particles, total angular momentum of the system is the (vector) sum of the angular momentum of the individual particles :

$$\vec{L} = \vec{l}_1 + \vec{l}_2 + \cdots + \vec{l}_n = \sum_{i=1}^n \vec{l}_i$$

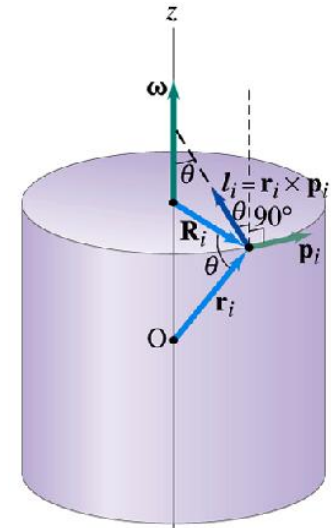
3. Angular momentum

Considering a rigid body rotates about a fixed axis --- ω axis.

The component of the total angular momentum along the rotation axis is

$$L = I\omega$$

I — the **moment of inertia** or **rotational inertia** on the axis.



Example

Four identical particles of mass m are mounted at equal intervals on a thin rod of length l and mass M , with one mass at each end of the rod. If the system is rotated with angular velocity ω about an axis perpendicular to the rod through one of the end masses, determine (a) the kinetic energy and (b) the angular momentum of the system.

Solution

- (a) The moment of inertia of the system about one end is

$$I = m(\ell/3)^2 + m(2\ell/3)^2 + m\ell^2 + (M\ell^2/3) = [(14m/9) + (M/3)]\ell^2.$$

The kinetic energy is

$$K = I\omega^2/2 = [(14m/9) + (M/3)]\ell^2\omega^2/2 = \boxed{[(7m/9) + (M/6)]\ell^2\omega^2}.$$

- (b) For the angular momentum we have

$$L = I\omega = \boxed{[(14m/9) + (M/3)]\ell^2\omega}.$$

4. Relation between angular momentum and torque

In linear motion, we have Newton's second law for a single particle:

$$\vec{F}_{\text{net}} = \frac{d\vec{p}}{dt},$$

(The time rate of change of **linear momentum of a particle** is equal to the **net force** applied to it.)

one can guess for its angular form


$$\vec{\tau}_{\text{net}} = \frac{d\vec{L}}{dt}.$$

The rotational equivalent of N-II law for **a particle**(or a system of particles, or a rigid body).

The time rate of change of **angular momentum** of a particle is equal to the **net external torque** applied to it.

For a rigid body, the component along the rotation axis is:

$$\sum \tau_{\text{axis}} = \frac{dL}{dt} = \frac{d}{dt}(I\omega) = I \frac{d\omega}{dt} = I\alpha$$

5. The conservation law of angular momentum

The time rate of change of **angular momentum** of a particle is equal to the **net external torque** applied to it.

$$\vec{\tau}_{\text{net}} = \frac{d\vec{L}}{dt}.$$

$$\text{If } \vec{\tau}_{\text{net}} = 0, \quad \vec{L} = \vec{C}$$

If **net external torque** acting on **a particle**(or particle's system, **or a rigid body**) is zero, the total **angular momentum** of the system remains **conserved**.

No exceptions to the law of conservation of angular momentum have ever been found.

Supplementary explanation:

1. It is equivalent to three component equations relating to the conservation of angular momentum in three mutually perpendicular directions (through the reference point)
2. It holds **beyond the limitation** of N's Laws (not only for particles whose speeds approach that of light, but also for subatomic particles).
3. **Condition:** $\vec{\tau}_{\text{net}} = 0$ **not** $\vec{F}_{\text{net}} = 0$. It is **only hold** for Inertia reference frame.
4. **For rigid body: Two cases** for fixed axis: (1) I keep constant, then ω is also constant, Same as self-rotation of the Earth; (2) I is variable quantity, such as particle or particle system. **As I decreases, ω increases, or its reverse.**

EXAMPLE 8.14 The Spinning Stool

GOAL Apply conservation of angular momentum to a simple system.

PROBLEM A student sits on a pivoted stool while holding a pair of weights. (See Fig. 8.32.) The stool is free to rotate about a vertical axis with negligible friction. The moment of inertia of student, weights, and stool is $2.25 \text{ kg} \cdot \text{m}^2$. The student is set in rotation with arms outstretched, making one complete turn every 1.26 s, arms outstretched. (a) What is the initial angular speed of the system? (b) As he rotates, he pulls the weights inward so that the new moment of inertia of the system (student, objects, and stool) becomes $1.80 \text{ kg} \cdot \text{m}^2$. What is the new angular speed of the system? (c) Find the work done by the student on the system while pulling in the weights. (Ignore energy lost through dissipation in his muscles.)

STRATEGY (a) The angular speed can be obtained from the frequency, which is the inverse of the period. (b) There are no external torques acting on the system, so the new angular speed can be found with the principle of conservation of angular momentum. (c) The work done on the system during this process is the same as the system's change in rotational kinetic energy.

SOLUTION

(a) Find the initial angular speed of the system.

Invert the period to get the frequency, and multiply by 2π :

$$\omega_i = 2\pi f = 2\pi/T = 4.99 \text{ rad/s}$$

(b) After he pulls the weights in, what's the system's new angular speed?

Equate the initial and final angular momenta of the system:

$$(1) \quad L_i = L_f \rightarrow I_i \omega_i = I_f \omega_f$$

Substitute and solve for the final angular speed ω_f :

$$(2) \quad (2.25 \text{ kg} \cdot \text{m}^2)(4.99 \text{ rad/s}) = (1.80 \text{ kg} \cdot \text{m}^2)\omega_f$$

$$\omega_f = 6.24 \text{ rad/s}$$

(c) Find the work the student does on the system.

Apply the work–energy theorem:

$$W_{\text{student}} = \Delta K_r = \frac{1}{2}I_f \omega_f^2 - \frac{1}{2}I_i \omega_i^2$$

$$= \frac{1}{2}(1.80 \text{ kg} \cdot \text{m}^2)(6.24 \text{ rad/s})^2$$

$$- \frac{1}{2}(2.25 \text{ kg} \cdot \text{m}^2)(4.99 \text{ rad/s})^2$$

$$W_{\text{student}} = 7.03 \text{ J}$$

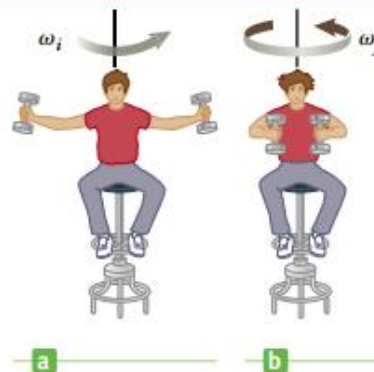


Figure 8.32 (Example 8.14) (a) The student is given an initial angular speed while holding two weights out. (b) The angular speed increases as the student draws the weights inwards.

He rotates at a relatively low speed when his arms are outstretched, but when she brings his arms in close to his body, he suddenly spins much faster. By remembering the definition of moment of inertia as $I = \sum mR^2$, it is clear that when he pulls his arms in closer to the axis of rotation, R is reduced for the arms so his moment of inertia is reduced. Since the angular momentum $I\omega$ remains constant (we ignore the small torque due to friction), if I decreases, then the angular velocity ω must increase.